

VIRTUALWORKS LAB INTERNSHIP PHARMACOVIGILANCE

TASK 5

INDIVIDUAL CASE SAFETY REPORT (ICSR)

Complete Pharmacovigilance Assessment of a Detailed Case Study

Submitted by	Hemraj Sonawane
Course / Program	VirtualWorks Lab Internship Pharmacovigilance
Task	Individual Case Safety Report (ICSR)
Date	29 August 2026

Academic integrity note: This report uses a fictional patient case created only for internship learning. It is not a real patient safety report, and no real patient identifiers are included.

1. Aim and Objective

Aim: To perform a complete pharmacovigilance assessment of a detailed adverse drug reaction case and prepare an Individual Case Safety Report (ICSR) in a Drug Safety Associate-style workflow.

Objective: To identify the suspected ADR, confirm ICSR validity, document patient and drug information, assess causality, seriousness, severity and expectedness, write a clear case narrative, and identify relevant follow-up information.

2. What is an ICSR?

An Individual Case Safety Report is a structured record of one or more adverse events/adverse drug reactions or other reportable observations in an individual patient. For regulatory submission, a valid ICSR normally requires four minimum elements: an adverse event/reaction, a suspect or interacting medicinal product, an identifiable patient, and an identifiable reporter.

Field	Details
Minimum element 1	At least one adverse event/adverse drug reaction or other reportable observation
Minimum element 2	At least one suspected or interacting medicinal product
Minimum element 3	An identifiable patient
Minimum element 4	An identifiable reporter

3. Detailed Case Study

A 46-year-old female patient (initials S.M.), weighing approximately 62 kg, was prescribed nitrofurantoin 100 mg orally twice daily on 11 August 2026 for an uncomplicated lower urinary tract infection. Her medical history included episodic migraine. She had no known chronic lung disease, no previous documented drug allergy and no recent hospital admission. Paracetamol 500 mg was used occasionally for headache.

On 19 August 2026, eight days after starting nitrofurantoin, she developed low-grade fever and a dry cough. Over the next two days she developed increasing shortness of breath and chest tightness. On 21 August 2026 she attended the emergency department. Oxygen saturation was 89% on room air, respiratory rate was 26/min and temperature was 38.2 C. Chest radiography showed bilateral interstitial opacities. Laboratory findings included white blood cell count $13.4 \times 10^9/L$ and eosinophils 8%. Influenza and SARS-CoV-2 tests were negative; blood cultures did not show bacterial growth.

Nitrofurantoin was stopped on 21 August 2026. The patient was admitted to hospital, given supplemental oxygen and short-course systemic corticosteroid treatment. Her breathing improved substantially within 48 hours. She was discharged on 24 August 2026 and was reported clinically recovered by 28 August 2026. Nitrofurantoin was not restarted.

4. ICSR Validity and Initial Triage

Field	Details
Case type	Initial spontaneous report from a healthcare professional (academic simulation)
Day zero	21 August 2026 - date the simulated safety unit received sufficient information to identify a valid case
Identifiable patient	Yes - female patient, initials S.M., age 46 years
Identifiable reporter	Yes - treating physician (fictional reporter)
Suspect medicinal product	Nitrofurantoin 100 mg
AE/ADR present	Yes - acute respiratory reaction with pneumonitis-type clinical picture
ICSR validity	VALID - all four minimum criteria are present

5. ADR Identification and Coding Approach

Field	Details
Reported symptoms	Dry cough, pyrexia, dyspnoea, chest tightness and reduced oxygen saturation
Clinical findings	Bilateral interstitial opacities, eosinophilia and hypoxaemia
Working diagnosis	Acute nitrofurantoin-associated pulmonary hypersensitivity / drug-induced pneumonitis
Suspected ADR	Acute pneumonitis / interstitial pulmonary reaction
Suggested MedDRA concept	Pneumonitis (illustrative training code - verify in the current licensed MedDRA browser)
Associated terms	Dyspnoea, cough, pyrexia and oxygen saturation decreased may be captured as additional terms if required by coding conventions

MedDRA note: In real case processing, the verbatim report should be coded using the current licensed MedDRA version and the organisation's coding conventions. The coding shown here is for internship practice only.

6. Suspected and Concomitant Medication

Medicine	Dose / Route	Frequency	Therapy dates	Indication	Role
Nitrofurantoin	100 mg oral	BD	11/08/2026 - 21/08/2026	Lower UTI	Suspect
Paracetamol	500 mg oral	PRN	Intermittent	Migraine/headache	Concomitant

7. Chronology of the Case

Date	Clinical event
11 Aug 2026	Nitrofurantoin 100 mg twice daily started.

19 Aug 2026	Dry cough and low-grade fever developed.
21 Aug 2026	Dyspnoea worsened; oxygen saturation 89%; hospital admission. Nitrofurantoin withdrawn.
22-23 Aug 2026	Improvement after oxygen and corticosteroid therapy.
24 Aug 2026	Patient discharged from hospital.
28 Aug 2026	Clinical recovery reported. No rechallenge performed.

8. Seriousness, Severity and Expectedness Assessment

Field	Details
Seriousness	SERIOUS
Seriousness criterion	Required inpatient hospitalisation
Severity / intensity	Severe - clinically important respiratory symptoms with hypoxaemia requiring oxygen and inpatient management
Outcome	Recovered
Expectedness	Expected for this academic exercise
Expectedness basis	The training reference safety information used for this case lists acute pulmonary hypersensitivity/pneumonitis-type reactions with nitrofurantoin. In real practice, expectedness must be checked against the applicable regional product label/reference safety information.

Seriousness and severity are not the same concept. This case is serious because hospital admission occurred. The reaction is also clinically severe because respiratory compromise required oxygen and active treatment.

9. Causality Assessment

9.1 WHO-UMC Assessment

Field	Details
Temporal relationship	Compatible - symptoms appeared after eight days of therapy
Known association	Compatible with a recognised nitrofurantoin pulmonary reaction
Dechallenge	Positive - clear improvement after withdrawal and treatment
Rechallenge	Not done; unnecessary and potentially unsafe
Alternative causes	Infectious causes were not completely impossible, but available tests and the response to withdrawal make them less likely
WHO-UMC category	Probable / Likely

9.2 Naranjo Algorithm (Educational Use)

Question considered	Score
---------------------	-------

Previous conclusive reports of this reaction?	+1
Event appeared after the suspected drug was given?	+2
Reaction improved after stopping the drug?	+1
Reaction reappeared on rechallenge?	0
Alternative causes reasonably excluded?	+2
Reaction reappeared with placebo?	0
Drug concentration in toxic range?	0
Dose-response evidence?	0
Objective evidence supporting the reaction?	+1
Total	7 - Probable

Overall causality conclusion: The timing, positive dechallenge, objective respiratory findings and lack of a stronger alternative explanation support a probable relationship between nitrofurantoin and the pulmonary reaction.

10. ICSR Case Narrative

A 46-year-old female patient (S.M.), weight approximately 62 kg, with a medical history of episodic migraine and no known chronic pulmonary disease, received nitrofurantoin 100 mg orally twice daily from 11 August 2026 for an uncomplicated lower urinary tract infection. Concomitant medication included occasional paracetamol 500 mg for headache. On 19 August 2026, approximately eight days after treatment initiation, the patient developed low-grade fever and dry cough. By 21 August 2026 she developed progressive dyspnoea and chest tightness and presented to the emergency department. At presentation, oxygen saturation was 89% on room air, respiratory rate was 26/min and temperature was 38.2 C. Chest radiography demonstrated bilateral interstitial opacities. Laboratory investigations showed white blood cell count $13.4 \times 10^9/L$ with eosinophils 8%. Testing for influenza and SARS-CoV-2 was negative, and blood cultures showed no bacterial growth. Nitrofurantoin was discontinued on 21 August 2026. The patient was hospitalised and treated with supplemental oxygen and a short course of systemic corticosteroid. Respiratory symptoms improved markedly within 48 hours. She was discharged on 24 August 2026 and was reported clinically recovered by 28 August 2026. Nitrofurantoin was not reintroduced. The event was assessed as serious due to hospitalisation, severe in intensity, and expected against the academic reference safety information used for this exercise. WHO-UMC causality was assessed as Probable/Likely and the educational Naranjo score was 7 (Probable).

11. Drug Safety Associate Review Checklist

Review item	Assessment
Case validity	Valid ICSR - four minimum criteria present
Duplicate check	No duplicate identified in this academic simulation
Patient details	Sufficient for case-level identification; no direct personal identifiers used
Suspect drug details	Drug, dose, route, frequency, indication and dates documented
Event information	Onset, clinical findings, treatment and outcome documented
Seriousness	Serious - hospitalisation
Expectedness	Expected for the stated academic RSI
Causality	Probable/Likely (WHO-UMC); Naranjo 7 = Probable
Narrative	Chronological and medically coherent

Follow-up required	Yes - useful additional details are listed below
--------------------	--

12. Follow-up Information I Would Request

- Exact product brand, manufacturer and batch/lot number, if available.
- Baseline and follow-up renal function, because renal impairment can influence nitrofurantoin exposure.
- Any previous nitrofurantoin exposure or history of similar respiratory symptoms.
- Smoking history, occupational exposures and previous pulmonary investigations.
- Detailed discharge diagnosis and final chest imaging result.
- Whether eosinophil count and oxygen saturation normalised after recovery.
- Any additional infectious or autoimmune investigations performed during admission.

13. Final ICSR Assessment Summary

Field	Details
Patient	S.M., 46-year-old female
Suspected drug	Nitrofurantoin 100 mg orally twice daily
Indication	Uncomplicated lower urinary tract infection
Primary suspected ADR	Acute pneumonitis / pulmonary hypersensitivity-type reaction
Time to onset	Approximately 8 days
Action taken	Nitrofurantoin withdrawn
Treatment	Hospital admission, supplemental oxygen and systemic corticosteroid
Outcome	Recovered
Seriousness	Serious - inpatient hospitalisation
Severity	Severe
Expectedness	Expected in the academic reference safety information
WHO-UMC causality	Probable / Likely
Naranjo score	7 - Probable
Rechallenge	Not performed
Case status	Valid initial ICSR; follow-up desirable for additional product and clinical details

14. Clinical and Pharmacovigilance Interpretation

The case contains a plausible drug-event time relationship and objective respiratory findings, with improvement after nitrofurantoin withdrawal. Hospitalisation makes the case serious, while the need for oxygen and active treatment makes the clinical intensity severe. The reaction is considered expected for this academic exercise, but in actual pharmacovigilance work expectedness must be determined using the current applicable regional product label or other approved reference safety information. The case would receive priority medical review because it is serious, and follow-up should focus on confirming diagnosis, excluding alternative causes and obtaining missing product details.

15. Conclusion

This task demonstrates the complete processing of a fictional Individual Case Safety Report from receipt to medical assessment. The four minimum ICSR criteria were confirmed, the suspected ADR was identified and documented, the case was classified as serious and severe, expectedness was evaluated, and causality was assessed as Probable/Likely. Writing the narrative and identifying targeted follow-up questions also reflects the practical responsibilities of a Drug Safety Associate.

16. References

1. International Council for Harmonisation (ICH). E2D(R1): Post-Approval Safety Data - Definitions and Standards for Management and Reporting of Individual Case Safety Reports. Final Step 4 guideline, adopted 15 September 2025.
2. International Council for Harmonisation (ICH). E2B(R3): Individual Case Safety Report (ICSR) Specification and Implementation Guide.
3. International Council for Harmonisation (ICH). E2A: Clinical Safety Data Management - Definitions and Standards for Expedited Reporting.
4. MedDRA MSSO. MedDRA Term Selection: Points to Consider. Current training guidance.
5. World Health Organization - Uppsala Monitoring Centre. The use of the WHO-UMC system for standardised case causality assessment.
6. Naranjo CA, et al. A method for estimating the probability of adverse drug reactions. Clin Pharmacol Ther. 1981;30(2):239-245.

Final note: This is an academic simulation for internship learning. Regulatory submission rules, timelines, coding conventions and expectedness assessments must follow the applicable company procedures and local/regional requirements in real pharmacovigilance practice.